

Lab 07

One-Dimensional Array in C++

Objective

The objective of this lab is to understand the concept of one-dimensional arrays in C++ and learn how to declare, initialize, and manipulate array elements.

Introduction

A one-dimensional array is a collection of elements of the same data type stored in contiguous memory locations. It allows storing multiple values in a single variable, making programs more efficient and organized.

Theory

In C++, an array is declared as:

```
data_type array_name[size];
```

Index starts from 0

- Elements can be accessed using index
- Arrays are useful for storing lists of data

Example

```
int arr[5] = {1, 2, 3, 4, 5};
```

Algorithm

1. Start
2. Declare an array
3. Take input from user
4. Store values in array
5. Display array elements
6. End

Program:

```
#include <iostream>
using namespace std;

int main() {
    // 5 elements ka array declare kar rahe hain
    int arr[5];

    // User se values input lena
    cout << "Enter 5 elements: ";
    for(int i = 0; i < 5; i++) {
        cin >> arr[i]; // har index par value store kar rahe hain
    }

    // Array ki values display karna
    cout << "Array elements are: ";
    for(int i = 0; i < 5; i++) {
        cout << arr[i] << " "; // har element ko print kar rahe hain
    }

    return 0;
}
```

Output:

User enters values and program displays them.

Lab Tasks

Task 01:

Array Declaration and Initialization

- Declare an integer array called "numbers" with a size of 5.
- Initialize the array with values 10, 20, 30, 40, and 50.
- Display the values of the array elements using a loop.

Code

```
#include <iostream>
using namespace std;

int main() {
    // Array ko values ke sath initialize kar rahe hain
    int numbers[5] = {10, 20, 30, 40, 50};

    // Array ke elements display karna
    cout << "Array elements are:\n";
    for(int i = 0; i < 5; i++) {
        cout << numbers[i] << endl; // har element ko alag line mein print kar rahe hain
    }

    return 0;
}
```

Task 02:

Accessing Array Elements

- Declare a character array called "message" to store a word of your choice.
- Use a loop to access each character of the array and display it on a separate line.

Code

```
#include <iostream>
using namespace std;

int main() {
    // Character array declare kar rahe hain jisme string store hai
    char message[] = "HELLO";

    // Har character ko alag line mein display karna
    cout << "Characters are:\n";
    for(int i = 0; message[i] != '\0'; i++) {
        cout << message[i] << endl; // jab tak null character na aaye, print karte raho
    }

    return 0;
}
```

Task 3:

Copying Arrays:

- Declare two integer arrays: "source" and "destination", each with a size of 5
- Initialize the "source" array with some values.
- Use a loop to copy the elements of the "source" array into the "destination" array
- Display the values of both arrays to verify the copy operation.

Code

```
#include <iostream>
using namespace std;

int main() {
    // Source array ko values ke sath initialize kar rahe hain
    int source[5] = {1, 2, 3, 4, 5};

    // Destination array declare kar rahe hain (empty hai)
    int destination[5];

    // Source se destination mein elements copy karna
    for(int i = 0; i < 5; i++) {
        destination[i] = source[i]; // har element copy ho raha hai
    }

    // Source array ko display karna
    cout << "Source Array:\n";
    for(int i = 0; i < 5; i++) {
        cout << source[i] << " "; // source ke elements print kar rahe hain
    }

    cout << endl;

    // Destination array ko display karna
    cout << "Destination Array:\n";
    for(int i = 0; i < 5; i++) {
        cout << destination[i] << " "; // destination ke elements print kar rahe hain
    }

    return 0;
}
```

Task 4:

Array Sum:

- Write a C++ program that prompts the user to enter 5 numbers.
- Store the numbers in an integer array.
- Calculate the sum of all the numbers in the array.
- Display the calculated sum.

Code:

```
#include <iostream>
using namespace std;

int main() {
    // 5 elements ka array declare kar rahe hain
    int arr[5];

    // Sum store karne ke liye variable initialize kar rahe hain
    int sum = 0;

    // User se 5 numbers input lena
    cout << "Enter 5 numbers:\n";
    for(int i = 0; i < 5; i++) {
        cin >> arr[i]; // har index par value store ho rahi hai
    }

    // Array ke tamam elements ka sum calculate karna
    for(int i = 0; i < 5; i++) {
        sum += arr[i]; // har element ko sum mein add kar rahe hain
    }

    // Final sum display karna
    cout << "Sum = " << sum;

    return 0;
}
```

Conclusion:

This lab helped in understanding how arrays work in C++ and how to efficiently store and access multiple values using a single variable.
